For the submission of your work:

- Create a folder named **RollNo_Name_DBI202_PaperNo**, e.g. se01245_LongNT_DBI202_01. **Do not** create any subfolder in this folder. All file created will be located in the above folder.

- For each question, you are required to write a database script. Create a file with the name corresponding to the index of the question. For example, **for question 1**, we will create a file named **Q1.sql** and create a file **Q2.sql for question 2**. So, if you do 10 questions, your folder must contain **only** 10 files Q1.sql, Q2.sql, Q3.sql, Q4.sql, Q5.sql, Q6.sql, Q7.sql, Q8.sql, Q9.sql and Q10.sql.

- Do not use any commands having the database name such as create database, alter database, use [database name], *etc*.

- Your response must contain only necessary commands for answering the question. Do not include any other command. For example, if you are required to create a trigger/procedure, then your response should contain only commands for creating the corresponding trigger/procedure; all commands for testing the created trigger/procedure are forbidden.

- Do not use GO command in your code

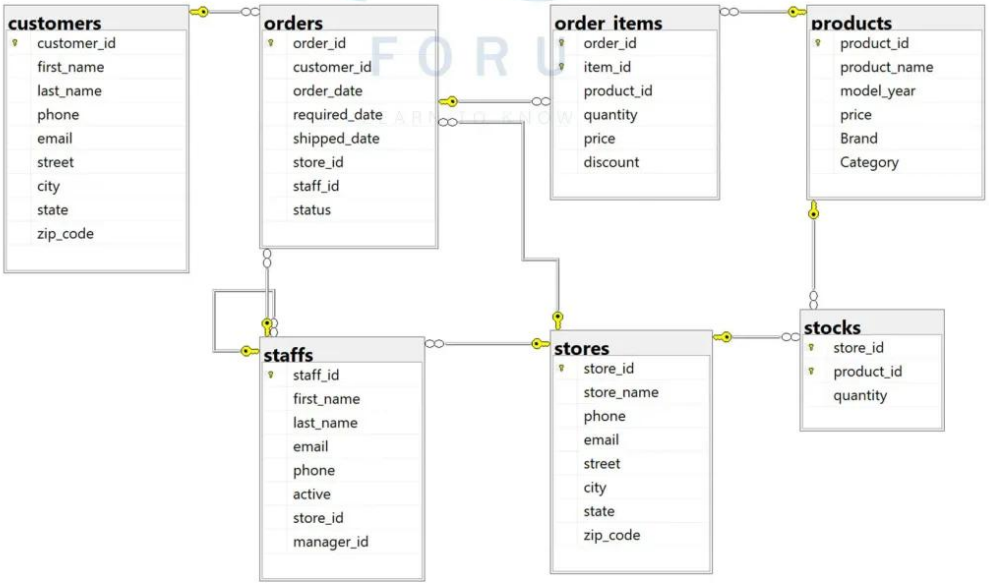
- On completion, import your work by browsing to the above folder.

- Note that:

+ You could use only SQL Server, SQL Server Management Studio, and paper or offline document in your computer.

+ If <u>any</u> of the previous requirements is not respected, your mark will be 0.

From the 2nd question, you should use the database provided in the .sql file which has the following database diagram. Please, run the provided script to create tables and insert data into your database.



Question 1:

Select all products having the model_year 2018 of the category 'Electric Bikes' as follows (1 mark):

	product_id	product_name	model_year	price	Brand	Category
1	185	Trek Conduit+ - 2018	2018	2799.99	Trek	Electric Bikes
2	186	Trek CrossRip+ - 2018	2018	4499.99	Trek	Electric Bikes
3	187	Trek Neko+ - 2018	2018	2799.99	Trek	Electric Bikes
4	188	Trek XM700+ Lowstep - 2018	2018	3499.99	Trek	Electric Bikes
5	189	Trek Lift+ Lowstep - 2018	2018	2799.99	Trek	Electric Bikes
6	190	Trek Dual Sport+ - 2018	2018	2799.99	Trek	Electric Bikes
7	191	Electra Loft Go! 8i - 2018	2018	2799.99	Electra	Electric Bikes
8	192	Electra Townie Go! 8i - 2017/2018	2018	2599.99	Electra	Electric Bikes
9	193	Trek Lift+ - 2018	2018	2799.99	Trek	Electric Bikes
10	194	Trek XM700+ - 2018	2018	3499.99	Trek	Electric Bikes
11	195	Electra Townie Go! 8i Ladies' - 2018	2018	2599.99	Electra	Electric Bikes
12	196	Trek Verve+ - 2018	2018	2299.99	Trek	Electric Bikes
13	197	Trek Verve+ Lowstep - 2018	2018	2299.99	Trek	Electric Bikes
14	198	Electra Townie Commute Go! - 2018	2018	2999.99	Electra	Electric Bikes
15	199	Electra Townie Commute Go! Ladies' - 2018	2018	2999.99	Electra	Electric Bikes
16	200	Trek Powerfly 5 - 2018	2018	3499.99	Trek	Electric Bikes

Picture 1.1

Question 2:

Write a query to select customer_id, first_name, last_name, city, state of all customers having the last_name begins with an ‘A’ or an ‘O’ and living in the states ‘CA’ or ‘NY’; display the results in ascending order of state then in ascending order of customer_id for customers from the same state as follows (1 mark):

	customer_id	first_name	last_name	city	state
1	30	Jamaal	Albert	Torrance	CA
2	530	Selene	Austin	Duarte	CA
3	538	Sherie	Ayala	South El Monte	CA
4	712	Kecia	Olsen	Santa Cruz	CA
5	767	Twana	Arnold	Anaheim	CA
6	1147	Kandace	Ayers	Santa Monica	CA
7	1175	Sindy	Anderson	Pomona	CA
8	1415	Harold	O'connor	Santa Cruz	CA
9	45	Bennett	Armstrong	Bethpage	NY
10	197	Andy	O'neill	Rego Park	NY
11	236	Verona	O'neill	Selden	NY
12	327	Sharie	Alvarez	New York	NY
13	366	Lavonne	Anderson	Floral Park	NY
14	378	Rosalba	O'neal	Mount Vernon	NY
15	381	Kristen	Alvarez	Bronx	NY
16	406	Kerrie	O'neill	Shirley	NY
17	499	Lidia	Ashley	Baldwinsville	NY
18	535	Inge	Olsen	Long Beach	NY
19	549	Elmo	Arnold	Long Beach	NY
20	563	Rodney	Odom	Richmond Hill	NY
21	616	Shery	Acosta	Saratoga Springs	NY
22	706	Lolita	O'neill	Massapequa Park	NY
23	721	Erlinda	Osborne	West Islip	NY
24	732	Lavona	Austin	Utica	NY
25	836	Corinna	Adams	Rosedale	NY
26	854	Phylis	Adkins	New York	NY
27	897	Bettyann	Acosta	Lancaster	NY
28	965	Margit	Osborn	Plainview	NY
29	992	Chere	Alston	Patchogue	NY
30	1123	Raphael	O'neil	Fairport	NY
31	1141	Mallie	Osborn	Newburgh	NY
32	1219	Alden	Atkinson	Shirley	NY
33	1222	Geraldine	O'donnell	Flushing	NY
34	1402	Sandee	Alvarado	Depew	NY
35	1406	Farrah	Orr	New Hyde Park	NY
36	1436	Macie	Ayers	Bellmore	NY

Picture 2.1

Question 3:

Write a query to display order_id, order_date, customer_id, customer_first_name, customer_last_name, store_id, store_name corresponding to orders of the 'Santa Cruz Bikes' store and the order_date of these orders is in June 2017 or in July 2017 as follows (1 mark):

	order_id	order_date	customer_id	customer_first_name	customer_last_name	store_id	store_name
1	926	2017-06-03	233	Mariana	Strong	1	Santa Cruz Bikes
2	927	2017-06-03	1438	Lee	Dunn	1	Santa Cruz Bikes
3	928	2017-06-05	957	Stephen	Vega	1	Santa Cruz Bikes
4	940	2017-06-12	585	Chasidy	Webster	1	Santa Cruz Bikes
5	941	2017-06-12	736	Genny	Hensley	1	Santa Cruz Bikes
6	944	2017-06-14	138	Jone	Bernard	1	Santa Cruz Bikes
7	947	2017-06-15	278	Myesha	Burgess	1	Santa Cruz Bikes
8	949	2017-06-16	867	Rosalva	Hamilton	1	Santa Cruz Bikes
9	958	2017-06-18	1276	Shona	Mcmillan	1	Santa Cruz Bikes
10	961	2017-06-19	82	Lizzie	Joyner	1	Santa Cruz Bikes
11	987	2017-07-02	172	Jeanie	Kirkland	1	Santa Cruz Bikes
12	991	2017-07-06	751	Hermila	Mckay	1	Santa Cruz Bikes
13	993	2017-07-09	1415	Harold	O'connor	1	Santa Cruz Bikes
14	995	2017-07-11	621	Basil	Ballard	1	Santa Cruz Bikes
15	996	2017-07-11	915	Beryl	Bennett	1	Santa Cruz Bikes

Picture 3.1

Question 4:

Write a query to display product_id, product_name, model_year, Brand, Category, TotalAmount corresponding to each product of the category 'Mountain Bikes' of the brand ‘Surly’, where TotalAmount of a given product is the total amount of money obtained from the sales of the corresponding product in all orders. Note that the amount obtained from the sale of a product in one order is calculated by (1-discount)*quantity*price. Display the results in descending order of TotalAmount, then in ascending order of product_name for rows having the same TotalAmount. Note that in the following figure, products with TotalAmount = NULL are products which have never been sold in any order (2 mark) .

	product_id	product_name	model_year	Brand	Category	TotalAmount
1	3	Surly Wednesday Frameset - 2016	2016	Surly	Mountain Bikes	102528.9747
2	6	Surly Ice Cream Truck Frameset - 2016	2016	Surly	Mountain Bikes	59326.8377
3	28	Surly Karate Monkey 27.5+ Frameset - 2017	2017	Surly	Mountain Bikes	44399.8224
4	36	Surly Troll Frameset - 2017	2017	Surly	Mountain Bikes	14877.2014
5	31	Surly Wednesday - 2017	2017	Surly	Mountain Bikes	14778.5595
6	30	Surly Ice Cream Truck Frameset - 2017	2017	Surly	Mountain Bikes	11359.8864
7	33	Surly Wednesday Frameset - 2017	2017	Surly	Mountain Bikes	8652.5159
8	27	Surly Big Dummy Frameset - 2017	2017	Surly	Mountain Bikes	7439.9256
9	126	Surly Big Fat Dummy Frameset - 2018	2018	Surly	Mountain Bikes	NULL
10	128	Surly ECR 27.5 - 2018	2018	Surly	Mountain Bikes	NULL
11	124	Surly Krampus - 2018	2018	Surly	Mountain Bikes	NULL
12	121	Surly Krampus Frameset - 2018	2018	Surly	Mountain Bikes	NULL
13	127	Surly Pack Rat Frameset - 2018	2018	Surly	Mountain Bikes	NULL
14	122	Surly Troll Frameset - 2018	2018	Surly	Mountain Bikes	NULL

Picture 4.1

Question 5:

Write a query to display staff_id, first_name, last_name, email, manager_id, store_id, store_name corresponding to all staff who has never made any order as in the following figure. Note that the email is the email of the staff (1 mark).

	staff_id	first_name	last_name	email	manager_id	store_id	store_name
1	1	Fabiola	Jackson	fabiola.jackson@bikes.shop	NULL	1	Santa Cruz Bikes
2	4	Virgie	Wiggins	virgie.wiggins@bikes.shop	2	1	Santa Cruz Bikes
3	5	Jannette	David	jannette.david@bikes.shop	1	2	Baldwin Bikes
4	10	Bernardine	Houston	bernardine.houston@bikes.shop	7	3	Rowlett Bikes

Picture 5.1

Question 6:

Create a stored procedure named p2 for calculating the number of orders of a given customer where customer_id int is the input parameter and NumberOfOrders int is the output parameter of the stored procedure (1 mark).

For example, when we execute the procedure p2 with customer_id = 5 by using the following statements, the result should be as in the following figure:

```
declare @x int

exec p2 5, @x output

select @x as NumberOfOrders
```

	NumberOfOrders
1	2

Picture 6.1

Question 7:

Create a trigger named T2 for the delete statement on table order_items so that when we call a delete statement to remove one or more rows from the table order_items, after deleting the corresponding rows, the system will display order_id, order_date, status, product_id, product_name, quantity corresponding to deleted rows (2 mark).

for example, when we execute the following statement, the results should be as in the following figure:

```
delete from order_items

where order_id = 1
```

	order_id	order_date	status	product_id	product_name	quantity
1	1	2016-01-01	Completed	4	Trek Fuel EX 8 29 - 2016	1
2	1	2016-01-01	Completed	16	Electra Townie Original 7D EQ - 2016	2
3	1	2016-01-01	Completed	10	Surly Straggler - 2016	2
4	1	2016-01-01	Completed	8	Trek Remedy 29 Carbon Frameset - 2016	2
5	1	2016-01-01	Completed	20	Electra Townie Original 7D EQ - Women's - 2016	1

Picture 7.1

Question 8:

Write a query insert an order with order_id = 2005, customer_id = 30, order_date is on 21 april 2023, required_date is on 26 april 2023, store_id = 2, staff_id = 5 and the status = 'Pending' (1 mark).

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Note: the retirement age for employees is from 30. When we execute this procedure using the statements as follow: exec proc_getRetiredEmployees 5, the result as follow:

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
Mark	U	Can	111222444	1993-05-12	379 Loop, Metal, TX	M	45000.00	333445555	5
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	35000.00	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000.00	888665555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	30000.00	333445555	5
Ahn	C	Seop	654321789	1988-11-07	456 Green, Houston, TX	M	80000.00	333445555	5
Ramesh	K	Naravan	666884444	1962-09-15	915 Fire Oak, Humble, TX	M	38000.00	333445555	5

Picture 8.1

Question 9:

[1 mark] Create a trigger named "tr_no_delete" for the delete statement on the Dependent table. This trigger does NOT allow you to delete any dependent

The system will display the result "no delete" if you execute the delete statement.

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for example, when you execute the following statement: delete from Dependent.

The system will display the result as in the following figure:

the result
no delete

Picture 9.1

Question 10:

[0.5] Write a query to increase employee salary by an additional 10% if he/she has at least 1 dependent under 18 years old.